Nassau Candy Distributor: Machine-Learning-Assisted Factory Reassignment for Shipping Lead-Time Reduction

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Abstract

This paper evaluates whether moving products between five candy factories can shorten shipping lead time for Nassau Candy Distributor, and finds that the simulation suggests an average 30.4% reduction but rests on assumptions the data cannot validate. We combine exploratory analysis of 10,194 order lines, three regression models that predict lead time, and a distance-based scenario simulator that scores every product, region, ship-mode and factory combination. The simulator produces 200 reassignment recommendations covering all 15 products and four regions, with a mean confidence score of 0.66. The regression models have negative R^2 (best: −0.006), so the recommendations come entirely from a geographic rule: ship from the factory nearest the region. Observed lead times also average about 1,300 days, which indicates a date-quality defect in the source file. We therefore present the results as a validated method and a hypothesis, not as a proven saving, and we list the checks required before any production change.

Keywords: supply-chain optimization, factory allocation, lead-time prediction, scenario simulation, regression, distribution analytics

1. Introduction

Nassau Candy Distributor sells 15 confectionery products across the United States and Canada, and each product is made at one of five fixed factories. Long delivery times raise cost, hurt customer satisfaction and tie up working capital. The project asks a practical question: if a product were produced at a different factory, how much shorter would its shipping lead time be, and what would that do to gross profit?

1.1 Objectives

1. Describe how lead time, sales and margin vary by region and ship mode.

2. Predict lead time from order attributes with machine-learning regressors.
3. Simulate moving each product to every other factory and rank the outcomes.
4. Report decision KPIs: lead-time reduction, profit impact, confidence and coverage.

1.2 Scope

The work covers one Python pipeline (`nassau_factory_optimization.py`, developed in the notebook `Pro_1_Final.ipynb`) and its Excel output with five sheets: EDA Summary, Model Metrics, Top Recommendations, KPI Summary and Route Distances. It was built as part of an internship task series. It does not model factory capacity, production cost, inventory or carrier contracts.

2. Data and Preprocessing

The input is a transaction file with 10,194 rows and 18 columns, one row per order line. Key fields are Order ID, Order Date, Ship Date, Ship Mode, Region, Division, Product Name, Sales, Units, Gross Profit and Cost. Four ship modes appear (Standard, Second, First Class and Same Day) and four regions (Atlantic, Interior, Pacific, Gulf).

2.1 Product-to-factory mapping

The file does not say where each product is made, so the project supplies a mapping of the 15 products to five factories.

Factory	Products	Location used
Lot's O' Nuts	Nutty Crunch Surprise, Fudge Mallows, Scrumdiddlyumptious	32.88, −111.77 (Arizona)
Wicked Choccy's	Milk Chocolate, Triple Dazzle Caramel	32.08, −81.09 (Georgia)
Sugar Shack	Laffy Taffy, SweeTARTS, Nerds, Fun Dip, Fizzy Lifting Drinks	48.12, −96.18 (Minnesota)
Secret Factory	Everlasting Gobstopper, Lickable Wallpaper, Wonka Gum	41.45, −90.57 (Iowa/Illinois border)
The Other Factory	Hair Toffee, Kazookles	35.12, −89.97 (Tennessee)

2.2 Derived fields

- **Lead Time Days** = Ship Date − Order Date, with dates parsed as day-month-year.
- **Profit Margin** = Gross Profit ÷ Sales.
- **Current Factory** from the mapping above.

2.3 Outlier handling

Rows outside the interquartile-range fence (Q1 − 1.5·IQR to Q3 + 1.5·IQR) are removed before modelling and before baseline lead times are computed. The exploratory tables use all rows.

2.4 Data-quality finding

Lead times in the source are not credible. Order dates fall in 2024–2025, but ship dates fall in 2026–2030, which gives a median lead time of about 1,270 days and a mean of 1,290–1,400 days in every region and ship-mode cell. Order IDs embed years such as 2021 and 2024, which suggests the two date columns were shifted by different amounts. [Likely] This is a data defect, not a logistics reality, and it affects every result that follows.

3. Methodology

The pipeline has four stages: exploratory analysis, predictive modelling, geographic scenario simulation and recommendation scoring.

3.1 Exploratory analysis

Orders, mean and median lead time, sales, gross profit and mean margin are aggregated by Region × Ship Mode (16 groups), and the lead-time distribution is plotted as a histogram.

3.2 Predictive models

Three regressors predict Lead Time Days from eight features: Product Name, Current Factory, Region, Ship Mode, Division (categorical) and Units, Sales, Gross Profit (numeric). Categorical features are imputed with the most frequent value and one-hot encoded; numeric features are median-imputed and standardized. Data are split 80/20 with a fixed random seed (42).

Model	Configuration
Linear Regression	Default
Random Forest	180 trees, minimum 3 samples per leaf
Gradient Boosting	160 trees, depth 3, learning rate 0.05

Models are compared on RMSE, MAE and R^2.

3.3 Distance model

Each region is represented by one reference city: Atlantic = New York area (40.0, −74.0), Interior = Chicago (41.9, −87.6), Pacific = Los Angeles (34.1, −118.2), Gulf = Houston (29.8, −95.4). The haversine great-circle distance from each factory to each region gives a 4 × 5 matrix of 20 routes.

3.4 Scenario simulation

For every product, region and ship mode, a baseline lead time is taken from historical data, using the most specific available average (product + region + ship mode, then product + region, then region + ship mode, then region). The lead time under a candidate factory is then scaled by relative distance:

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T_{cand} = T_{base} \left( \frac{d_{cand}}{d_{current}} \right)^{0.35}
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The exponent 0.35 is a chosen constant, not an estimated one. It encodes the assumption that lead time grows less than proportionally with distance, because fixed handling time does not shrink.

3.5 Scoring and ranking

- **Lead-time reduction %** = 100 × (T_base − T_cand) ÷ T_base.
- **Estimated profit impact** = average gross profit per order × reduction fraction × 0.15, where 0.15 is an assumed pass-through of speed to profit.
- **Confidence score** = 0.55 + 0.35 × min(orders ÷ 1,000, 1), capped at 0.90, so it measures sample size only.
- **Recommendation score** = reduction % + 0.002 × profit impact.

For each product, region and ship mode, the highest-scoring candidate that differs from the current factory is kept as the top recommendation.

4. Results

The predictive models learn nothing useful, while the scenario simulator yields a 30.4% mean lead-time reduction across 200 recommendations.

4.1 Exploratory findings

Across the 16 Region × Ship Mode groups, average lead time ranges only from 1,287 days (Gulf, Same Day) to 1,397 days (Interior, Same Day), and medians sit between 1,269 and 1,274 days. Same Day shipping is no meaningfully faster than Standard Class (gaps of under about 80 days on a 1,300-day base, and Same Day is slower in the Interior and Pacific regions), which is not plausible for real shipments and supports the data-quality finding in Section 2.4. Margins are stable at roughly 66–67% in every group. Standard Class in Pacific (1,617 orders), Atlantic (1,487) and Interior (1,230) carries the most volume.

4.2 Model performance

Model	RMSE (days)	MAE (days)	R^2
Linear Regression	266.80	215.30	−0.0065
Gradient Boosting	267.71	216.00	−0.0134
Random Forest	273.03	221.63	−0.0541

All three models score at or below zero R^2 on the held-out 20%, so none beats predicting the mean lead time. Linear regression is nominally best, but its edge is noise. The features (product, factory, region, ship mode, units, money values) carry no signal about the lead-time column as it stands.

4.3 Reassignment recommendations

The simulator keeps 200 top recommendations: 50 product-region pairs × 4 ship modes. Ten of 60 product-region pairs already use the nearest factory, so they receive no recommendation. Every recommendation moves a product to the factory closest to the region's reference city.

Region	Move from → to	Products	Mean reduction (%)	Mean profit impact per order
Interior	Lot's O' Nuts → Secret Factory	3	54.25	0.75
Pacific	Wicked Choccy's → Lot's O' Nuts	2	45.40	0.59
Interior	Wicked Choccy's → Secret Factory	2	42.72	0.56
Pacific	The Other Factory → Lot's O' Nuts	2	39.56	0.47
Pacific	Secret Factory → Lot's O' Nuts	3	39.33	1.55
Pacific	Sugar Shack → Lot's O' Nuts	5	38.03	0.20
Interior	Sugar Shack → Secret Factory	5	37.55	0.20
Atlantic	Lot's O' Nuts → Wicked Choccy's	3	33.21	0.46
Interior	The Other Factory → Secret Factory	2	32.77	0.39
Gulf	Sugar Shack → The Other Factory	5	28.55	0.15
Gulf	Lot's O' Nuts → The Other Factory	3	22.06	0.31
Atlantic	Sugar Shack → Wicked Choccy's	5	18.93	0.10
Gulf	Wicked Choccy's → The Other Factory	2	18.24	0.24
Gulf	Secret Factory → The Other Factory	3	17.81	0.70

Region	Move from → to	Products	Mean reduction (%)	Mean profit impact per order
Atlantic	The Other Factory → Wicked Choccy's	2	10.80	0.13
Atlantic	Secret Factory → Wicked Choccy's	3	8.56	0.34

The largest gain, 54.25%, comes from moving the three Lot's O' Nuts chocolate bars (Arizona) to Secret Factory for Interior customers, because the distance falls from about 2,348 km to 251 km. Reduction percentages are identical for all products and ship modes on the same route, since the formula depends only on the distance ratio.

4.4 Decision KPIs

KPI	Value	How it is computed
Lead Time Reduction %	30.40	Mean over 200 recommendations
Profit Impact Stability	0.533	1 − (std ÷ mean) of profit impact, clipped to 0–1
Scenario Confidence Score	0.663	Mean sample-size confidence
Recommendation Coverage	1.000	Products with a recommendation ÷ all products (15 of 15)

Estimated profit impact is small: mean 0.41 per order (median 0.25, maximum 2.47). The confidence score is 0.90 for the five Wonka Bars with 1,800–2,100 orders each, but only about 0.55 for products with fewer than 10 orders such as Hair Toffee (4 orders) and Everlasting Gobstopper (3 orders).

5. Discussion and Limitations

The pipeline is sound as a decision-support method, but the 30.4% figure cannot be treated as a forecast. Six issues limit it.

1. **Corrupted lead times.** A 1,300-day baseline is impossible, so every baseline, and every day saved, is inflated. Percentages survive rescaling; absolute days do not.

2. **The simulator is a distance rule, not a learned model.** The regressors have R^2 below zero and are never used in the simulation. Reduction % is fully determined by 1 − (d_cand ÷ d_cur)^0.35, so "nearest factory wins" is built in, not discovered.
3. **Unvalidated constants.** The exponent 0.35 and the 0.15 profit pass-through are assumptions. Changing them changes every number, and no sensitivity analysis is run.
4. **Coarse geography.** Each region is one city, although Interior and Pacific span thousands of kilometres of customers. Straight-line distance ignores carriers, ports and routes.
5. **No cost or capacity side.** Moving a product requires equipment, ingredients, labour and capacity at the new factory. The profit estimate counts only a speed benefit on existing gross profit, with no relocation cost.
6. **Confidence is sample size only.** A score of 0.90 means "many orders", not "likely to be correct". It says nothing about model accuracy.

The profit-impact result is also small. At 0.41 per order on average, even large lead-time gains imply little financial benefit under the stated assumptions, so factory moves would need justification beyond margin, such as service levels or customer retention.

A smaller point: because the recommendation maps each region to its nearest factory, products from different factories converge on the same few sites (for example, all Pacific recommendations point to Lot's O' Nuts). Real capacity at those sites could become a bottleneck.

6. Recommendations and Future Work

Fix the data before acting on any recommendation, then pilot the highest-volume moves.

6.1 Before any decision

1. Repair the Ship Date and Order Date columns at source, or obtain the original 2021-era dates, and recompute lead times. Expect realistic values of a few days.
2. Re-run the models. If R^2 remains near zero on clean data, drop the regressors or add real drivers such as carrier, distance, weight and weekday.
3. Replace the fixed exponent 0.35 with a coefficient fitted from clean lead time against route distance, and report its confidence interval.
4. Run a sensitivity analysis on the exponent and the 0.15 profit factor.

6.2 Pilot candidates (subject to the checks above)

- **Interior orders for the three Lot's O' Nuts bars** moved to Secret Factory: largest simulated gain (54%) and about 5,700 order lines for these three bars across all regions.
- **Pacific orders for Wicked Choccy's bars** moved to Lot's O' Nuts: 45% simulated gain on bars with about 4,150 order lines across all regions.
- **Low-volume products** (fewer than 10 orders) should be excluded from decisions until more data exists.

6.3 Model improvements

- Use customer city or postal code instead of four regional points, and road or rail distance instead of great-circle distance.
- Add factory capacity, changeover cost and transport cost, and turn the ranking into a constrained optimization, such as an assignment or linear program.
- Replace the sample-size confidence score with a prediction or bootstrap interval.
- Validate with a backtest or a controlled pilot on one product-region pair.

7. Conclusion

The project delivers a complete, reproducible workflow for testing factory reassignment, and its main result is a ranked list of 200 moves with an average simulated lead-time reduction of 30.4%. That result is a geographic hypothesis: the predictive models show no skill, the key constants are assumed, and the lead-time data contain a clear date error. The workflow is worth keeping. Its numbers should be treated as provisional until the data are repaired, the distance relationship is estimated from clean data, and a pilot confirms the effect.